

PLANT BREEBING

1. Plant introduction

Plant introduction is the movement of new varieties from one environment into another of similar characteristics, within a country or from other regions.

Types of plant introduction.

a) Based on adaptation

i.Primary introduction: Variety is well adapted to the new environment, released for commercial cultivation without any alteration in the original genotype.

ii.Secondary introduction: Introduced variety may be subjected to selection and hybridization to isolate a superior variety.

b) Based on utilisation

i. Direct introduction: New variety takes no time for establishment.

ii. Indirect introduction: New variety takes some time for establishment

Objectives of plant introduction.

To get new sources of resistance against both biotic and abiotic stresses.

To enrich the germplasm (living genetic resources such as seed or another plant part) collecting e.g. groundnuts.

To introduce high yielding varieties to increase food production.

To introduce new plant species thereby creating ways to build up new industries.

• Advantages of plant introduction.

It protects the variability from genetic erosion by collecting germplasm.

It provides superior crop by introducing into new disease free areas.

• Disadvantages of plant introduction.

Introduction of noxious weeds e.g. water hyacinth.

Introduction of disease

Introduction of pests, e.g. potato tuber moth from Italy.

Ornamentals turned weeds, e.g. Lantana camara.

Threat to ecological balance, e.g. Eucalyptus species introduced from Australia

2.Plant selection

Selection in plant breeding

- Selection is a process of allowing certain plants to be parents of future generations.
- The chosen plants are supposed to be carriers of desirable genes that make them produce better.
- The rejected ones are supposed to be carriers of undesirable genes that depress their productivity

Types of plant selection

1.Natural selection

It is a process that operates in nature without human interference. Plants that survive through the adversities of nature are preferred and the weaker are wiped out (Survival of the fittest). natural selection has given the cultivated crops and "ecotype" in plants

2.Artificial selection

- Eye is the powerful tool in artificial selection.
- Man decides on which crop individuals to use in the next generation.
- Farmer looks at characteristics like yield, quality, disease tolerance, and height.
- An understanding of botanical characteristics is needed.

Methods of artificial selection

(i) Mass selection.

- This is a selection of a number of phenotypically superior plant heads or seed from the field population.
- It is based on phenotype (external character) and the harvested seeds are composited with progeny testing.
- Most vigorous plants from the mixed population of a crop are selected.

(ii) Progeny selection.

- This is commonly used in cross pollination, and often cross pollinated crops.
- Progeny (offspring) is a new individual organism that results from the process of sexual or asexual reproduction.

(iii) Pure-line selection.

- Pure-line selection is a process of selecting a desirable homozygous individual from the mixed population and multiply the same without contamination to release a new individual.

(iv) Clonal selection.

- A clone is a variety propagated vegetative from a single plant.
- Clonal selection is the selection and propagation of the desirable variations between the clones as well as within the clone.
- It is practical with non-flowering plants and those species which produce seed poorly.

3) Hybridization

Hybridization refers to the natural or artificial process of combining varieties or species of crops to create a hybrid.

* **A hybrid is an** offspring of two plants of different varieties, with characteristics that are better than either of their parents.

* **Heterosis or** hybrid vigour is the increase in growth, size, fecundity, or yield in the offspring.

Objectives of hybridisation

- To combine the desired characters into a single individual.
- To exploit and utilize the hybrid varieties.
- To artificially create a variable population for the selection of types with desired combination characters.

Types of hybridization

1. Intraspecific or inter-varietal hybridization: • The crosses are made between the plants belonging to two different varieties.

2. Intra-varietal hybridization: • The crosses are made between the plants of the same variety.

3. Interspecific or intragenic hybridization: • The crosses are made between two different species of the same genus.

4. Introgressive hybridization: • The transfer of some genes from one species into the genome of the other species.

Advantages and disadvantages of hybridization.

Advantages

- Crossing two parental individuals gives chance for formation of new genetic recombinants and increase viability.
- Widened selection base exploits hybrid vigour or heterosis.
- Offspring with better characteristics are produced.
- The characteristics include, drought resistance, growth rate and improved yield.

Disadvantages

- Seed is expensive.
- Seed production requires expertise.
- Seed production takes a long time.
- New seed must be bought for each planting season, to maintain high yields..

Procedure of hybridisation

It involves the following steps:

i. Selection of parents. Parental plants are selected from the local areas and are supposed to be best suited to the existing conditions.

ii. Selfing of parents or artificial self-pollination. It is essential for inducing homozygosity for eliminating the undesirable characters and obtaining inbreds.

iii. Emasculation. Inbreds are grown under normal conditions and are emasculated. Emasculation is the removal of stamens from female parent before they burst and shed their pollens. Emasculation is not essential for unisexual plants.

iv. Bagging. The emasculated flower or inflorescence is immediately bagged to avoid pollination by any foreign pollen. The bags are tied to the base of the inflorescence or to the stalk of the flower with the help of a thread.

v. Tagging. Tags with information of the emasculated flower are attached at the base of the flower.

vi. Crossing. Mature, fertile and viable pollens from the male parent are placed on the receptive stigma of emasculated flowers to bring about fertilisation. Pollen grains are collected in paper bags (maize) or petri dishes (wheat) and applied to the receptive stigmas with a camel hair brush, tooth pick or forceps.

vii. Harvesting and storing the F1 seeds. Crossed heads or pods of desirable plants are harvested. They are threshed or shelled after complete drying. Seed are stored with the original tags.

viii. Raising the F1 generation. Stored seeds are sown separately to raise the F1 generation. The plants of the F1 generation are hybrids.

Hybridization Methods of Plant Breeding in Cross Pollinated crops

It involves crossing of two or more inbreds. These crosses are single, double and three way hybrids.

i. Single cross hybrid

- Two inbred varieties (A and B) are crossed to produce a monohybrid (single cross AB).

ii. Three-way cross hybrid

- To produce a tri-hybrid or three-way hybrid, a monohybrid (AB) is crossed with an inbred variety (C).

iii. Double cross hybrid.

- Crossing two monohybrids (AB x CD) produces a di-hybrid or double hybrid.

Open Pollinated Varieties (OPVs)

An **OPV** variety is one whose seed is produced by random cross pollination (that is there is no pollination control).

The pollination of the plants in the field is not controlled, which means the crop will not be uniform, for example the crop will vary in plant height, the colour of silks will vary, the cobs will not be the same size and shape and the plants will mature at different times

Advantages of growing OPV

1. Low or no seed cost. The producer can keep part of the crop for seed.
2. The cost of seed is a lot less than hybrid seed.
3. More money can be spent on buying fertiliser or pesticide.
4. Seed can be recycled. That is grain from this season can be planted next season.
5. Low potential areas cannot justify the high cost of hybrid seed.
6. OPV have a broader genetic base and are more variable in flowering dates.

This results in a longer flowering period, which will enable an OPV to pollinate during short periods of high stress (for example moisture stress, temperature, etc.). This variation can at times, offer more stable yields than more uniformly flowering hybrids.

7. It is important to buy certified seed every three years to maintain genetic purity.

Disadvantages of OPV:

1. The yield potential is typically 10 - 25% less.

2. In high potential areas, OPV's will reduce profit margins.
3. They will not be uniform in colour, maturity and other plant characteristics.
4. Could impact on the price of the grain, i.e. quality.
5. Lack of uniformity may lead to difficulties in carrying out certain operations, such as spraying and harvesting, (especially when using a combine harvester).
6. To keep an OPV pure, it should be planted at least 300 m from other varieties.
7. Poor seed quality (seed kept for planting of the next crop is usually stored in poor conditions and is exposed to high temperature, pests and diseases) can result in poor germination and weak plants that cannot compete well with weeds.
8. OPV's are not genetically modified to exhibit insect or herbicide resistance.

Hybrids Advantages of growing hybrid maize are:

1. Hybrids are generally higher yielding than open pollinated varieties.
2. Hybrids are uniform in colour, maturity and other plant characteristics which enables the producer to carry out certain operations (for example fertilising, spraying and harvesting) at the same time.
3. The uniformity of the grain harvested can also have marketing advantages when sold to buyers with quality standards.
4. Hybrid yield on average 18% more than OPV.

Disadvantages of growing hybrids:

1. Seed is expensive.
2. The producer needs to yield more than two ton per hectare to justify the higher cost of seed.
3. New seed needs to be purchased every planting season.
4. The grain produced from hybrid seed may not be used as seed for the following planting seasons.
5. The producer might not be able to source seed timeously.
6. Hybrids are more susceptible to stress conditions (for example tasselling).
7. Under poor crop management and harsh environmental conditions, the yield advantage of hybrids over OPV's is diminished.

TUTORIAL QUESTIONS

1. Explain plant introduction as a breeding method.
2. Describe types of plant introduction.

- 3. Discuss advantages and disadvantages of plant introduction.**
- 4. Explain selection as a method of plant breeding.**
- 5. Describe types of selection in plant breeding.**
- 6. Discuss the advantages and disadvantages of plant selection.**
- 7. Explain methods of hybridization**
- 8. Describe advantages and disadvantages of hybridization.**
- 9. Describe hybrid seed production.**